

Anticipate biology.

Before it changes the
balance of power.

An evidence-driven research engine
for scientists working on what comes next.



A warning about the way we innovate.

“The United States tends to play catch-up after critical technologies have already become mainstream.”

National Security Commission on Emerging Biotechnology

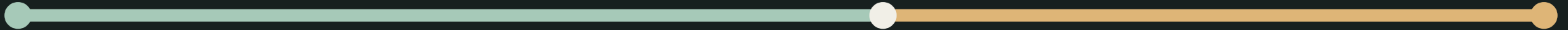
2025 FINAL REPORT / SECTION 1.3

A biological lead can become a unilateral advantage.

Novel biological threats leave less room for catch-up.

Capability develops

Capability becomes visible



ANTICIPATE

REACT

Connect evidence. Test implications.
Prepare defensive options earlier.

Response starts with
less lead time.

TEAM THESIS · Conceptual timeline, not measured intervals. Reactive measures remain necessary.

Point compute at the questions that matter.

Published evidence

- testable connections
- experiments worth reviewing



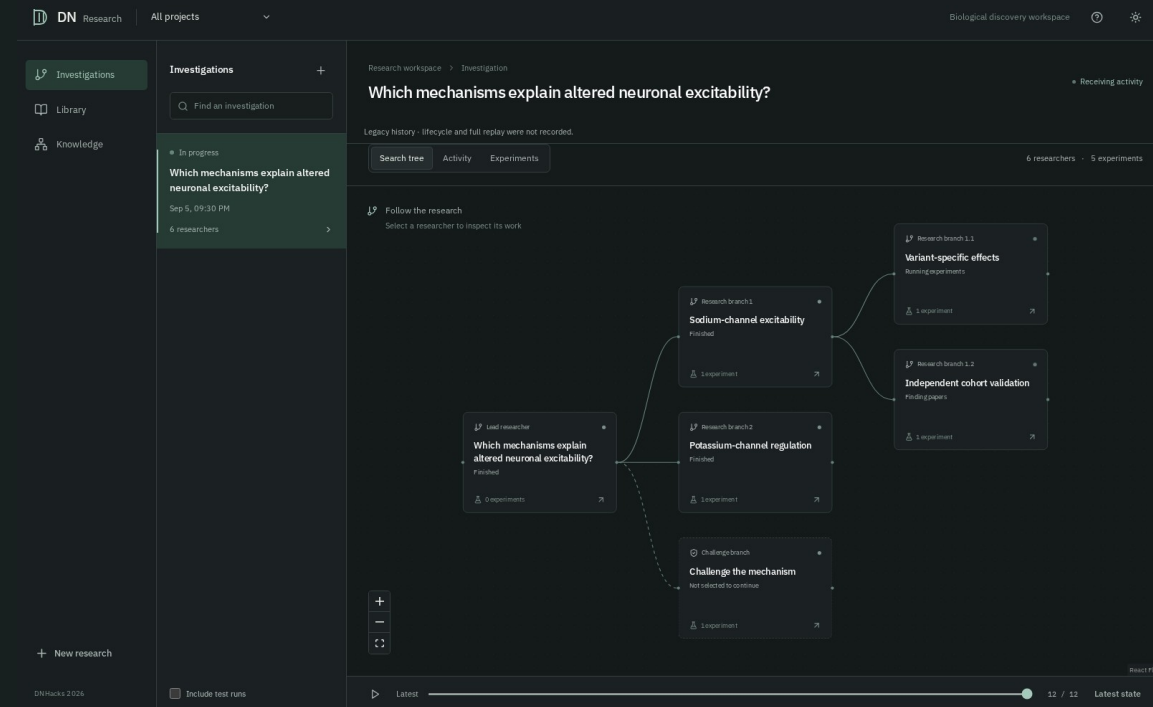
INITIAL PROVING GROUND · Computational biology, including pancreatic-cancer research.

Make the next experiment a deliberate decision.

Computational biologist

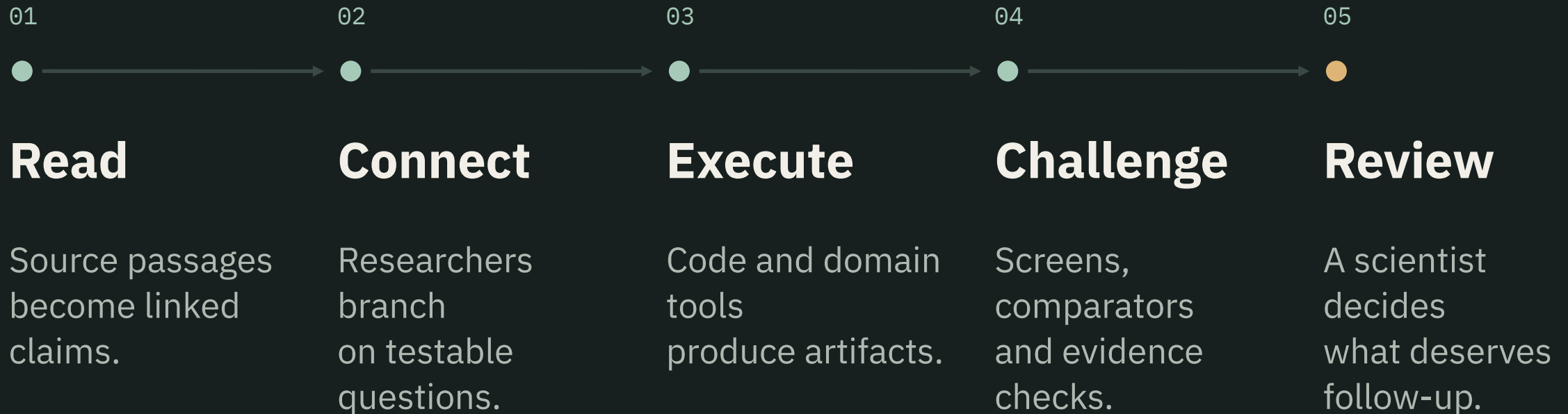
Small research team

Choose a question.
Inspect its evidence.
Explain the decision.



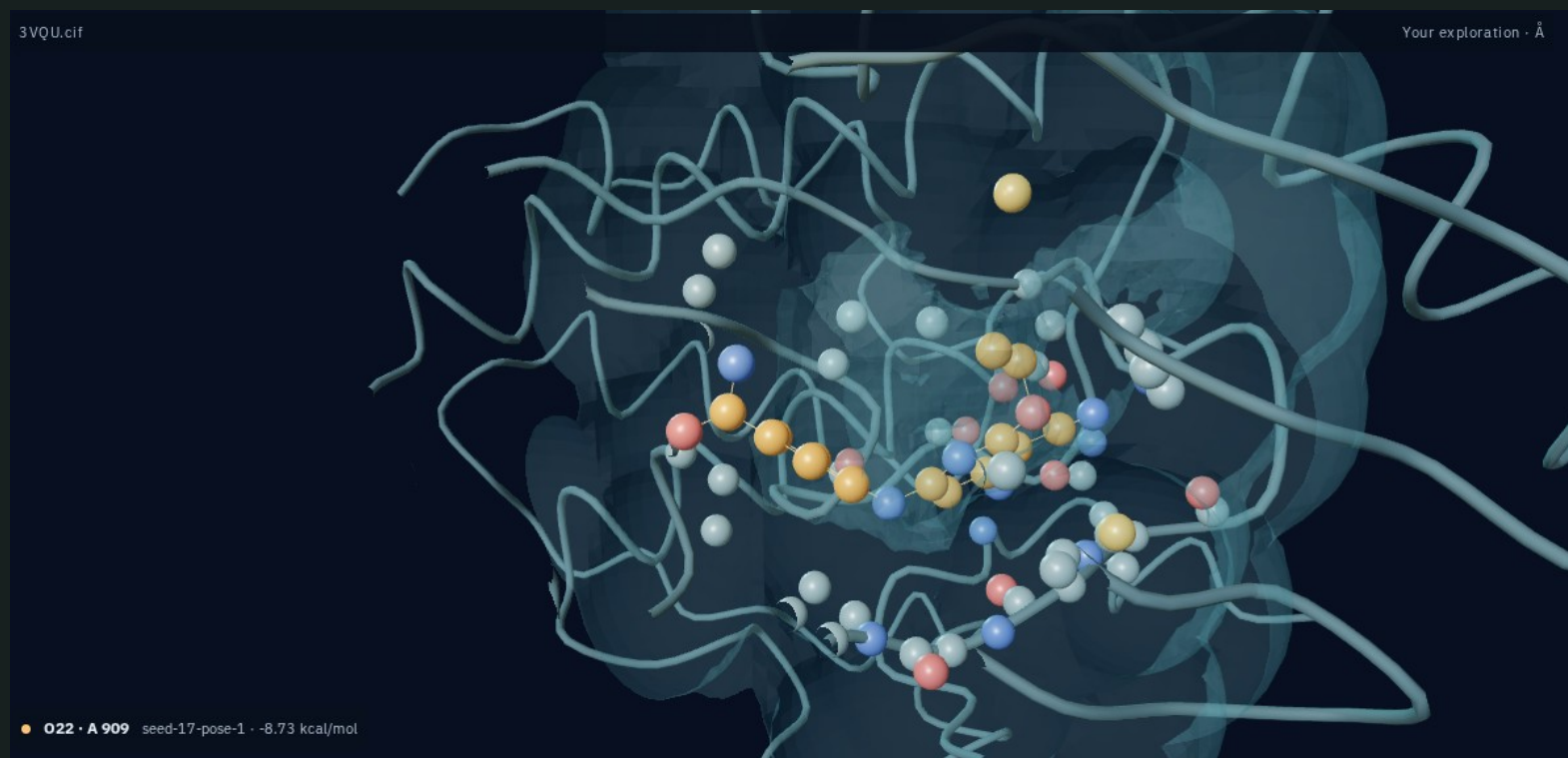
USER HYPOTHESIS · Actual interface with synthetic test data; interviews and usability study remain ahead.

The graph directs exploration. Experiments constrain it.



AI IS FUNDAMENTAL · Question formation, branching, executable analyses and artifact interpretation.

Inspect the image. Check the coordinates.



A real molecular scene

Image critique

Request a cutaway

2.714 Å coordinate check

ACTUAL 3D WORKBENCH · Image-to-coordinate review; the separate docking recovery benchmark did not pass.

Training is real. Advantage still has to be earned.

PROTEIN

219 / 110 / 105

train / validation / development

PCA beats the denoiser
on hidden-value
reconstruction.

DRUG RESPONSE

245 / 40 / 71

train / validation / test groups

PCA selected on validation.
No pancreatic-subset
advantage.

CELL ECOSYSTEMS

45 / 11 / 16

train / validation / external donors

Exact PCA remains best
for external reconstruction.

MEASURED DEVELOPMENT · Different tasks and units. These counts are not pooled or a discovery score.

Evidence needs a budget for being wrong.

INTERNAL E-VALUE TOOLS

1.12%

ever crossed the threshold

112 / 10,000 synthetic null runs
95% Wilson interval: 0.93–1.35%

**Prevent
leakage**

Train on past data; score fresh observations.

**Keep
receipts**

Freeze inputs, consume donors once, preserve replay.

**Respect
limits**

Synthetic calibration is not discovery accuracy.

SEPARATE DIAGNOSTIC · Learned encoder; 96 synthetic pairs/run. No investigation-wide guarantee.

Follow one question all the way to a reviewable result.

01 / SOURCE

**Verbatim claim +
passage**

04 / RESULT

Raw output + units

02 / BRIDGE

**Agent's concise
decision**

05 / VERIFICATION

Checks + failed controls

03 / EXPERIMENT

Saved code + dataset

06 / REVIEW

**Recorded human
decision**

REVEAL STORYBOARD · No qualifying run attached. Hidden from the default judge pitch.

First, show what survived.

CANDIDATE 01

**Claim in plain
language**

Result • comparator • scope

CANDIDATE 02

**Claim in plain
language**

Result • comparator • scope

CANDIDATE 03

**Claim in plain
language**

Result • comparator • scope

UNPOPULATED STORYBOARD • Card count is a layout example, not a measured survivor count.

Then reveal where the candidate leads.

The surviving candidate

Connect the measured claim to the later published finding.

LATER DISCOVERY PAPER

**Verified paper
cover goes here**

Title • authors • date • DOI
Exact match to the candidate

UNVERIFIED REVEAL SLOT • A matching paper has not been selected. Do not present as a discovery.

What could the system actually know?

Accessible corpus	100 curated full-text papers; publication years 2007–2025.
Curation boundary	Inclusive 2026-01-25 boundary; bind the actual run manifest.
Excluded evidence	Prove the revealed paper was excluded from every accessible source.
Remaining limitation	A frozen corpus does not eliminate model prior knowledge.

CURATION FACTS $\neq$ VERIFIED RUN BOUNDARY · Corpus, tool access and model-memory audit still required.

Start with one research team. Earn the right to expand.

Initial buyer	A research lead choosing computational experiments for a small team.
First product	A private research workspace, connected to approved data and tools.
Commercial hypothesis	Workspace subscription plus metered compute; validate willingness to pay.
Pilot acceptance	Useful reviewed hypotheses, independent reruns, total cost and analyst time.

PROPOSED BUSINESS MODEL · No customer, pricing, revenue or cost-saving claims.

The next milestone is proof that survives outside the demo.

REPRODUCE

One bounded investigation

Freeze task and baselines.
Independent scientist rerun.

HARDEN

One private pilot

Separate identities and data.
Constrain access and spend.

VALIDATE

One useful workflow

Measure quality and cost.
Test the user workflow.

RESEARCH-USE ROADMAP · Clinical use, regulated deployment and operational threat forecasting are not validated.

Give scientists a head start.

On the biology that will shape
industrial strength, health and security.

Our ambition: reduce the window
of unilateral biological advantage.



Every judging criterion has an evidence trail.

MAIN PRIZES

25 Problem / impact

SLIDES 1-4,16

50 Technical execution

SLIDES 6-13,18-24,28-29

25 Feasibility / deployment

SLIDES 14-15,26

BEST USE OF AI

25 Novelty / importance

SLIDES 4,6,25

50 Technical sophistication

SLIDES 6-9,18-24,28-29

25 Reliability / evaluation

SLIDES 9-13,19-24,28-29

BEST IN DESIGN

30 Design / craft

SLIDES 5,7,18

40 UX / usability

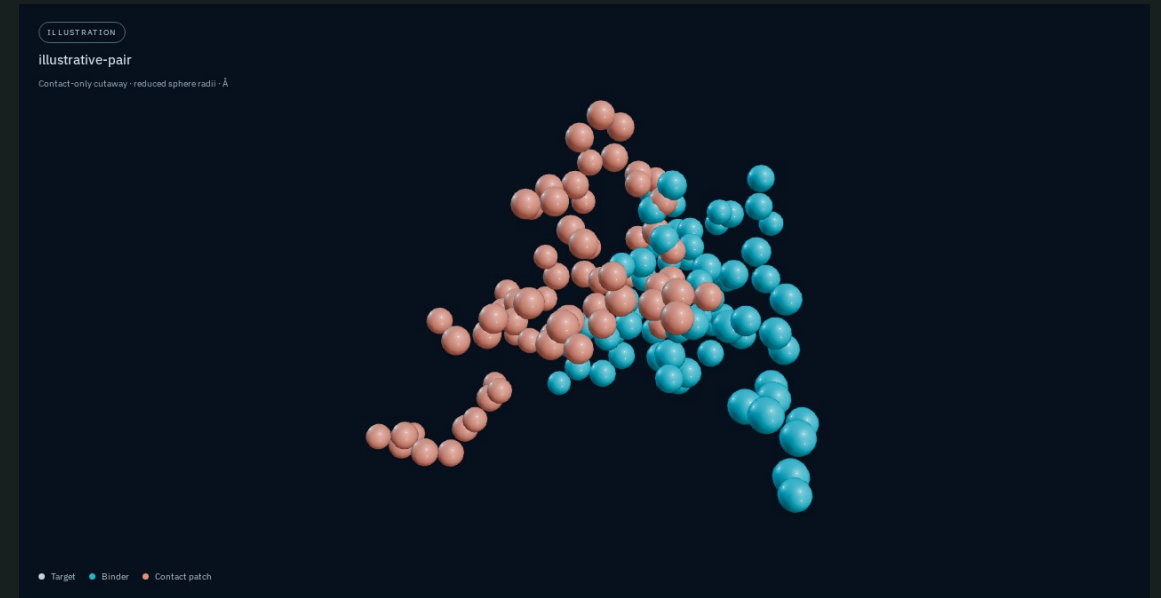
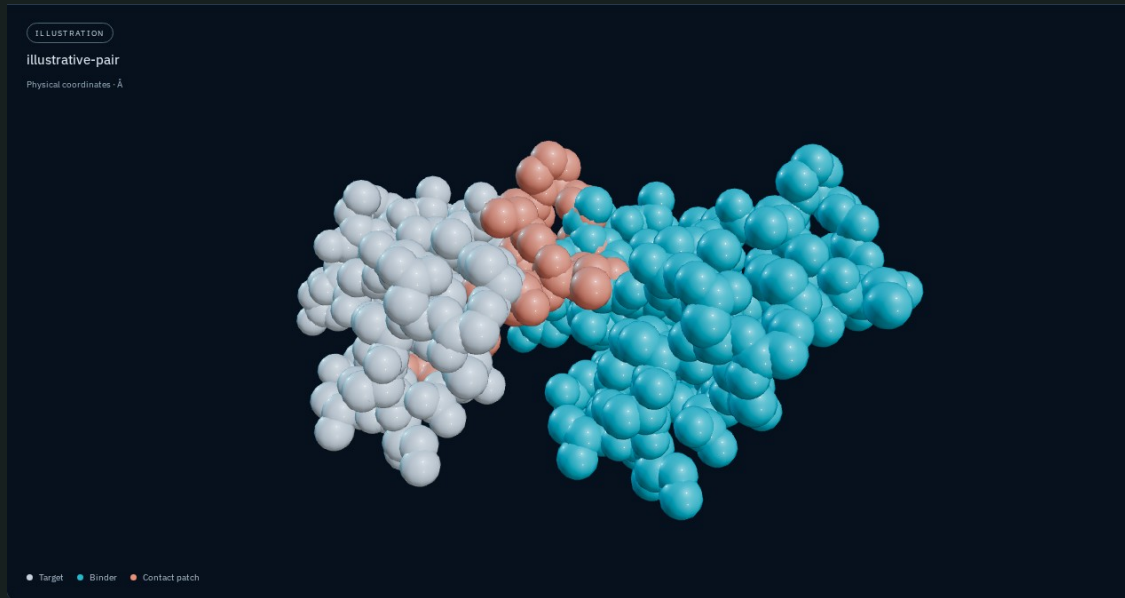
SLIDES 5-7,18

30 Understanding the user

SLIDES 5,14-15

Weights supplied by the team · No score or award outcome predicted.

A useful view makes a hidden problem visible.



Orient in the complete structure

Inspect the contact-only cutaway

ACTUAL UI CAPTURES · Same illustrative geometry. Color is paired with labels; opposing views expose occlusion.

What we trained—and what we did not.

Expression	Masked autoencoder + sequential neural better	Reconstruct expression; score fresh paired observations.
Protein	PCA / rank-PCA / masked denoiser	Learn protein-abundance structure and missingness.
Drug response	Separate encoders + ridge + bounded critic	Predict response; test association between two views.
Cell ecosystems	Cell encoders + lineage model + donor critics	Represent populations; predict labels; compare donor views.
Branch monitor	Per-prefix logistic models: implementation only	Real-trajectory fitting and calibration still pending.

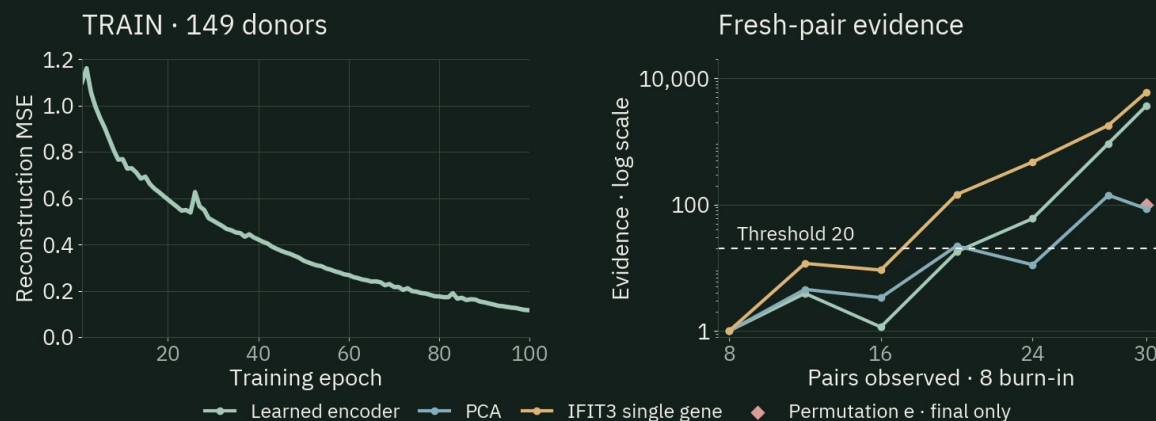
Pretrained LLMs / embeddings are dependencies. Renderers and fixed statistical tests are not team-trained models.

A measured training curve.

A separate evidence trajectory.

Learning the representation; measuring the evidence

Measured expression diagnostic · COVID cohort · no per-epoch validation history



Same 30 selected evaluation pairs. IFIT3 is strongest in this primary run; no PDAC result.

Source: real-expression/{training,evaluation}.json · measured loss / primary evidence paths

374 day-zero donors

TRAIN 149 / DEV 74 / EVAL 151
 19,000→512→128 encoder
 15% masking · 100 epochs
 Masked reconstruction objective

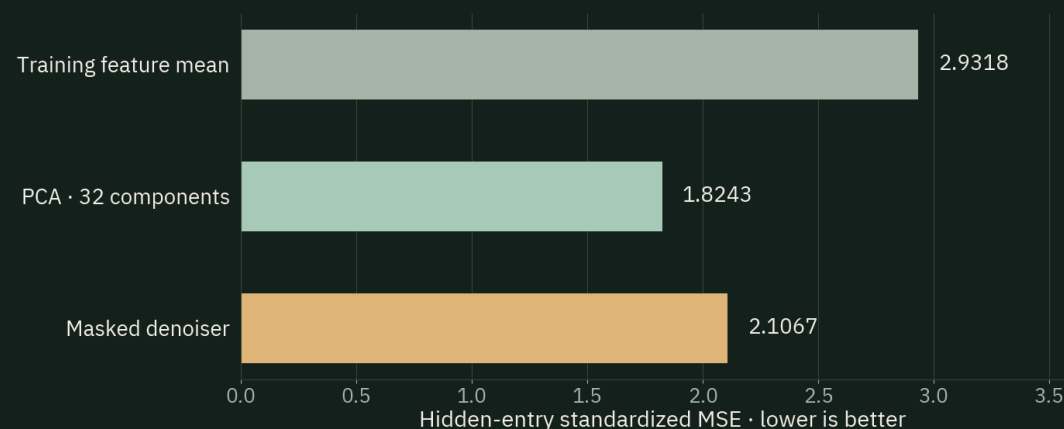
Single-gene baseline has greater final evidence in the primary run.

GSE212041 · Observational COVID cohort; not PDAC. Training loss only; no epoch-level validation history retained.

The protein baseline won the held-out reconstruction test.

The protein denoiser does not beat PCA

Measured hidden-entry reconstruction · CPTAC · 219 TRAIN / 110 LUAD validation cases



Final held-out error, not an epoch curve. No reconstruction history was retained.

Source: docs/protein-training.md · protein-s10-20260906-v2 · frozen reconstruction table

219 train / 110 validation

CPTAC: breast + colon train
Lung cohort validates
105 pancreatic tumors:
development set

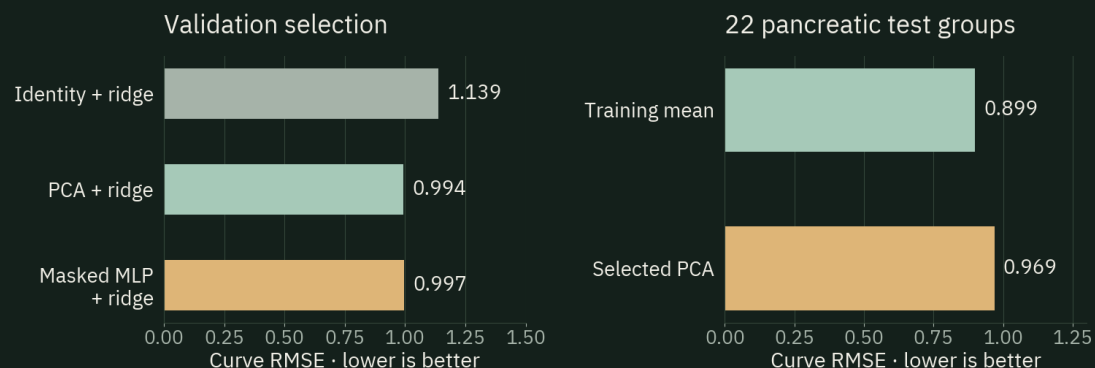
Denoiser: 4000→256→32→2000
20% corruption · 50 epochs
Hidden-value reconstruction

protein-s10-20260906-v2 · Standardized hidden-entry MSE, lower is better. Epoch loss histories were not retained.

Two real datasets. Two limits on the claim.

Model selection is not proof of pancreatic utility

Measured PRISM / CCLE development · 245 train / 40 validation / 71 test

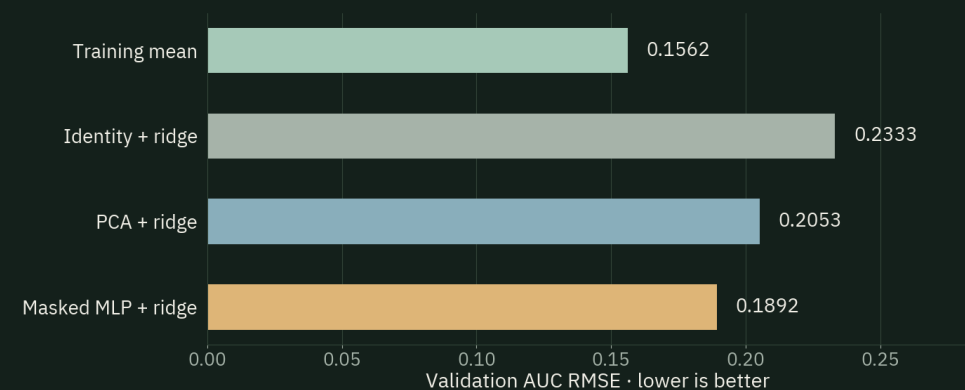


PCA wins model selection; the mean baseline is better on pancreatic test groups.

Source: docs/pharmacotype-cuda-training.json · trials / heldout.pancreatic_development

On real pancreatic organoids, retain the mean baseline

Measured Shi 2022 AUC benchmark · 21 train / 5 validation / 12 test patients



No learned model improves validation error. Five validation patients limit the conclusion.

Source: docs/pharmacotype-pdo-auc.json · mean_baseline / trials

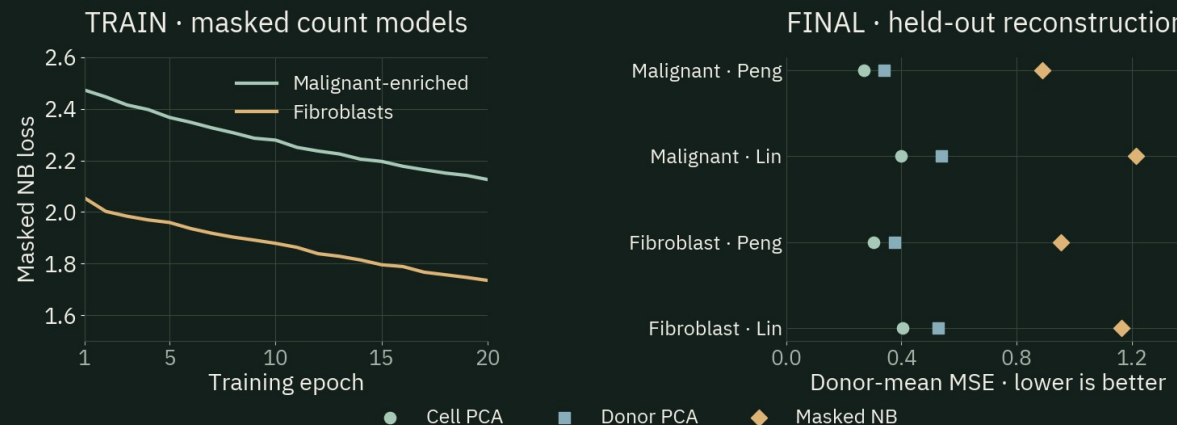
PRISM: 245/40/71 groups · PDO: 21/5/12 donors. Lower RMSE is better. These endpoints and scales differ.

Represent the cells.

Keep the donor as the unit of evidence.

A falling loss is only the first check

Measured cell-compartment models · training histories and separate final held-out errors



Cell PCA wins every shown held-out comparison. No per-epoch validation history recorded.

Source: docs/ecosystem-training/metrics.json · models.*.manifest.training_loss / held_out

Peng → Lin transfer

16 training donors
8 held-out donor partition
32 cells per eligible donor
2,000 genes per compartment

Cell PCA retained.
Neural gain unproven.

INITIAL PILOT · 20 NB training epochs + final held-out errors. Expanded train/validation curves: slide 29.

Internal e-value tools and branch monitoring answer different questions.

Biological tools

Is a specified measured relationship supported under its sampling contract?

Fixed statistics

Permutation / calibrated tests are statistical logic, not trained encoders.

Branch monitor

Could this research branch produce a qualifying result under a frozen budget?

Current readiness

Monitor code exists. Real-trajectory training and calibration remain uncompleted.

NO LIVE CALIBRATED PRUNING · Learned branch ratios are not automatically exact e-processes.

The differentiation has to be demonstrated.

Existing research agents	Elicit supports research workflows; DeepMind Co-scientist generates and evaluates hypotheses.
Our proposed wedge	Executable investigations with specialized biological tools, receipts and an inspectable tree.
The comparison to run	Strong single agent, flat parallel agents and our controller under the same total budget.
The outcome that matters	Independent rerun success, expert usefulness, time and total cost—including failures.

POSITIONING HYPOTHESIS · No first-in-category, superior-performance or defensible-moat claim yet.

What a pilot must settle.

Cost + infrastructure

Meter model calls, jobs, storage and review. Benchmark a complete investigation.

Security + integration

Enforce service identities, approved sources, restricted execution, audit export and access controls.

Regulation + adoption

Start with research use; assess dataset licenses, privacy, clinical scope and user workflow.

Proof boundary

No verified later-paper recovery, clinical benefit, customer traction or production certification.

Evidence snapshot: fee3165 · Appendices are for questions; check the manifest before presenting.

Every 3D tool keeps its own evidence boundary.

Binder workbench

Recorded scene, image observation and reverse view. Illustrated geometry; affinity unmeasured.

Molecular geometry

Scoped preparation, docking, image review and coordinate checks. Recovery benchmark did not pass.

Spindle mechanics

Pinned Cytosim CPU worker; real pilot exports 11 three-dimensional frames. Calibration remains ahead.

Tumor–stroma

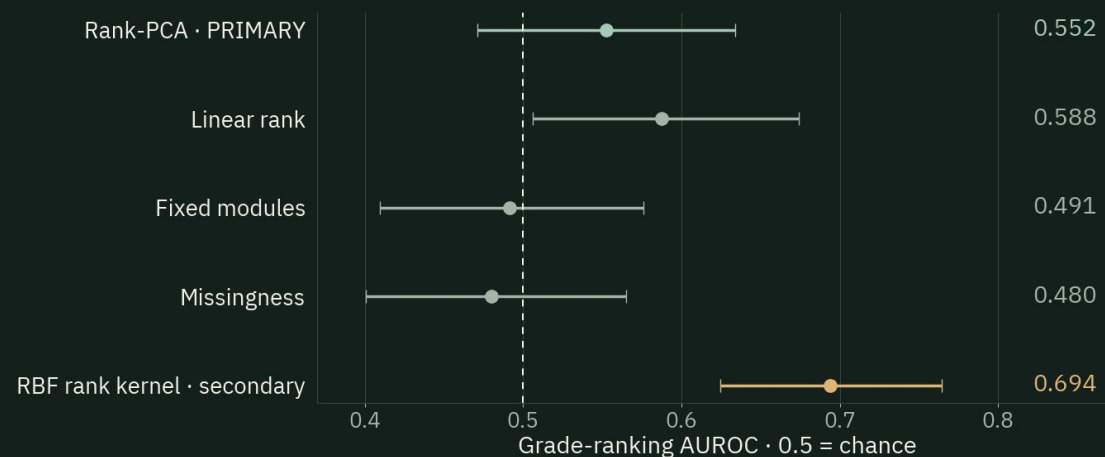
Native simulation and image workflow are active development beyond the audited snapshot.

3D geometry, simulated mechanics and biological experiments are different evidence types.

External data changed the conclusion.

A real external benchmark, with mixed results

Fudan PDAC · 224 graded tumors / 65 possible grade pairs · 95% bootstrap intervals



Cross-assay exploratory benchmark. Primary transfer is weak; secondary result needs a new cohort.

Source: docs/protein-external-{results,training}.json · all_graded / bootstrap_ci95

224 graded tumors

Fudan external benchmark
159 G3 / 65 G2 tumors
Five prespecified views

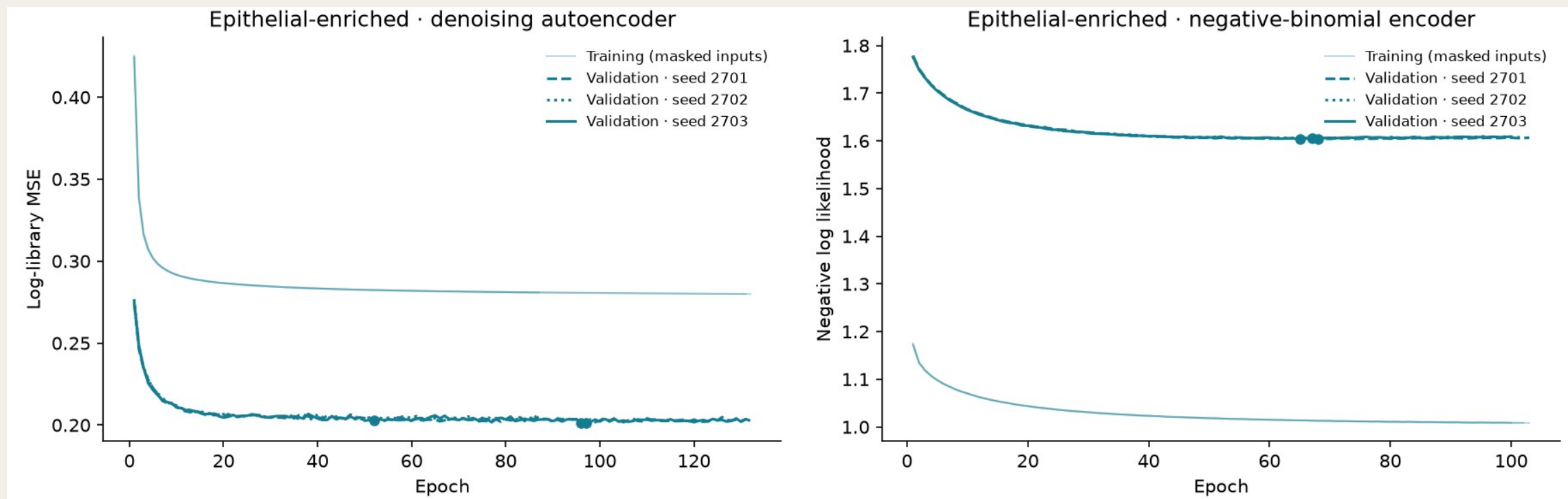
**Weak primary transfer.
Secondary signal needs
independent evaluation.**

CROSS-ASSAY EXPLORATORY BENCHMARK · Bootstrap intervals condition on the frozen model and observed cases.

Real validation curves.

PCA still leads externally.

72 DONOR RECORDS / 45 TRAIN / 11 VALIDATION / 16 EXTERNAL



EPITHELIAL MODELS SHOWN · Actual three-seed histories. Full fibroblast curves and source JSON are linked in Notes.